Machine Learning Application for Image Forgery Detection

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***Abstract*- We have proposed a novel machine learning based algorithm for the detection of image forgery without the availability of any prior information. Computation of local features is performed from the morphological information of image histograms and used to extract discriminating parameters. Images are assumed to be characterized by different parameters in different regions. Such parameters are learned by the image processing application with machine learning algorithms itself together on the segmentation in genuine and doubtful parts. A proposed supervised algorithm results on a wide range of images are very encouraging and proved to be sufficient for reliable forgery detection.**

Index Terms—Image processing, forgery detectionand, local image descriptors, histogram.

I. INTRODUCTION

Image authenticity is important in everyday life and everyone's life due to the advent of electronic social media networks and conventional media networks such as newspaper, magazines and television [1] etc. Bylooking any picture, it is not easy task to decide whether it is original picture or forged one. Every image is not required to check for authenticity but some of the cases like criminal investigation process and if editor of the press want to publish any forged image under the political pressure or some monetary gain then he/she may require to check the authenticity of the published picture (i.e. original image or forged image) [2]-[4].

For this purpose picture phony detection techniques are required, by which it can be identify that the received pictures are authentic or forged one [5]-[8]. According to the recent studies and research, picture phony detection techniques are classified into two categories i.e. active methods and non-active methods (or passive methods) [9]-[12]. In the active method, some kind of hidden or un-hidden information is inserted along with the picture at the time of capturing from the camera such as image signature, visible or hidden watermark. In non-active or passive method, some information or data is not inserted inside a picture at the time of capturing in form of image signature and watermarks, it deals with only binary information of the picture for features extraction and to check its authenticity. The copy-create and copy-move are the common method to develop forged picture, using these method some important or relevant information can hide from the picture and claim it is genuine picture. The forged area is modified with the help of edge blurring operation to get the presence of genuine picture. The goal of the proposed work is to develop computationally efficient passive techniques for identifying phonies in computerized pictures and efficient algorithms are proposed for the techniques to improve its detection accuracy and reduce computational complexity.

Visual digital data has become increasingly importantin recent years as a source of information. Social media platforms enable the quick and easy sharing of images and videos among a wide number of people. According to statistics [1], a sizable fraction of the world's population is in possession of a digital camera and is capable of taking images. Additionally, one-third of thepopulation has access to the internet [2] and can submitphotos to websites and social media. In this way, every user has the potential to be a source of information that is shared through the visual information he offers.

II. RELATED WORK

M. Derich [17], M. Ali Qureshi, and others Procedure for copy move fraud place. The DCT blocks and DWT are represented using this manner. It offers the FMT (Fourier Millen change), which allows the picture's highlights to be divided into a square. Strong to pivot, scaling, rotating, commotion expansion, and JPEG pressure are among the highlights. Adjacent squares and counters now include etymology arranging.

A detachable criminological approach was provided by Shashank Sharma et al. [18] for locating an assault on advanced recordings that was duplicated. The method of video falsification is divided into two categories: dynamic and latent. It depends on the existence of the initial video or on advanced watermarking. The video that provides the casing is an MPEG design with a pixel-by-pixel flow. The cycle is in clever innovation to recognize spatial, transient fabrication which is interrelated to the field of bury and intra outline movement to distinguish outline addition and cancellation, the strategy needs a grayscale picture so RGB picture that changed over to a grayscale picture.

R. Dhanya, R. Kalai Selvi, and others outlined a plan for the top-of-the-line investigation of duplicate move fabrication identification techniques. Using techniques like the Gaussian motion detection method, pressure on

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revolution, scaling, re-examining, and other methods inthis vector computation by applying head part modification, one can generally say that this strategy is hearty for some (PCT). The imitation identification method that deconstructs the square likeness into k-d tree representations suggests breaking up the initial image into random, no overlapping square fragments.

Ruhi Garg, Navpreet Kaur Gill, and others Computerized picture imitation location processes are the suggested tactic. They diminished the issues of picture verification in numerous areas like clinical sciences, organizations, and the phony discovery, which decides the validity of picture that ought to be credible. The interaction relies upon two sorts that is dynamic and aloof. The strategy summed up outline of the picture imitation is to be distinguished in square and central issue based methods. One of the significant strategy Brute power technique is utilized, the picture cycle is compacted in JPEG design which decides the goes through the antiquities of picture fabrication.

Wing administrator Nimit Kaura [21] et al. presented the investigation if SIFT and SURF highlights for duplicate, move picture fraud location. This strategy is to distinguish and find the phony recognition methods that are classified in two distinct structures. The first is central issue and another is block based fraud recognition. In which they fundamentally utilized (SIFT, SURF, DWT, DCT and SVD) calculations to recognize the fraud which somewhat filling the Gaussian role in the picture. This strategy functions admirably, regardless of whether the picture is incredibly compacted.

III. Methodology

Convolutional Neural Networks (CNNs or ConvNets) are a kind of neural networks that function well for image categorization and recognition. These neural networks are particularly good at identifying objects, faces, and traffic signs. Additionally, they support autonomous driving in remote-controlled cars and robot vision [18]. The LeNet architecture, the original CNN that was helpful in the creation of deep learning, served as their ancestor [19]. Every CNN is built on thefollowing four operations:

1) A picture made up of a pixel value matrix: Every image can be represented as pixel value matrices using computer graphics theory. Channels are a name for specific visual elements. Red, green, and blue are the three channels that make up digital camera photos. [20]. Each of these three channels has a pixel value in the range of 0 to 255, and they are stacked in layers to produce a 2-dimensional matrix. The existence of a single channel distinguishes grayscale images from other types.

2) The step of convolution: Extraction of features from the input image is required for the convolution step. With the aid of tiny squares ofinput data that are used to learn the properties of the image, this technique preserves the patterns between the pixels. The process is moving a single channel matrix, like orange, by one pixel over the base image, which might be green. A stride is the term used to describe this sliding step. Every position is multiplied element-by-element. The final integer, which denotes a single component of the desired output matrix, such as pink, is created by adding these products. A filter and a convolved feature interact with the input picture duringthe operations in this block to extract features from it.

3) Non-linearity (rectified linear unity [ReLU]): The ReLu includes a non-linear operation after each convolution process. Every pixel is affected by the ReLu, which changes all negative pixel values in the feature map to zero. Since practically every datum learned in the CNN has a linear feature, the ReLu seeksto introduce non-linearity to the CNN.

4) The combining phase: The feature map's most important information is kept while being reduced in dimension by the pooling stage through spatial pooling [21]. The pooling process involves combining many types, including average, total, and maximum. For instance, if the procedure calls for maximum pooling, the spatial neighbourhood must first be established before choosing the largest element from the rectified feature map inside the window [21]. For average pooling and sum pooling, the operation utilises the average rather than the greatest element. Pooling, convolution, and ReLU are thus the fundamental building elements for a CNN implementation that works.

Image Forgery Detection Algorithm:

1. Start
2. Input Image -- % using filename , pathnameSelect an Image from Dataset;
3. read input image :'Name','Input Image;
4. display input image
5. image resize at[300,300]);
6. gray scale conversion
7. Histogram Plot
8. apply canny edge to detect the boundary
9. apply logic for segmentation
10. load Training data
11. getting train data
12. resize train data
13. convert the train data and input into grayscale
14. subtracting the each pixel from train data tosplicing
15. creating initial binary image for approx location ad pixel values
16. Gray scale to binary image conversion ;

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17. show binary image
18. segregating the splicing coordinates from the original pixels
19. imshow(double(BW))
20. applying the morphological operation ofimage opening
21. contour detection
22. boundary detection
23. for loop 1 to number of boundary plottingboundary of splicing
24. Show Detected Region
25. splicing region indication
26. applying color for the spliced region
27. Calculation of Performance Measures -- %
28. Precision, recall, fmeasure values are stored in the variable "Performance"
29. plotting the performance value as table
30. end

IV. Result And Discussion

Figure 1. Input Image

The above mentioned algorithm is applied on different images. Figure 1 shows an image taken as input for detection of forgery .This is colored image in RGB space. The given image is an 8 bit 2D array in RGB plane. The image is resize at [300 300] and passed through the gray scale conversion.

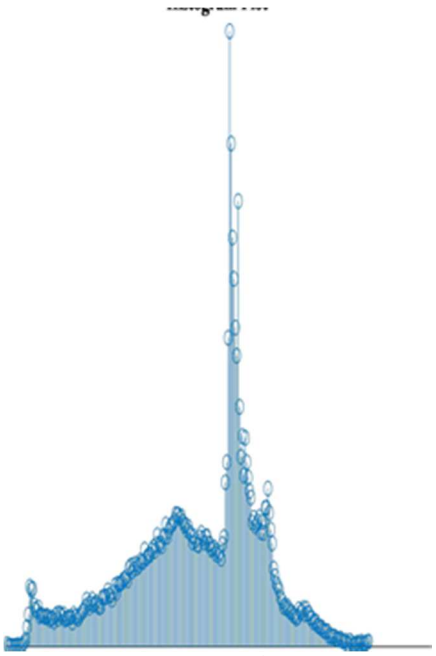

Figure 2. Image Histogram

Figure 2 is the image histogram obtained from the gray scale image. The histogram is representing the gray scale level in x axis and the range of x axis is from 0 to 255.The y axis shows the number of pixels associated to a given gray scale level.

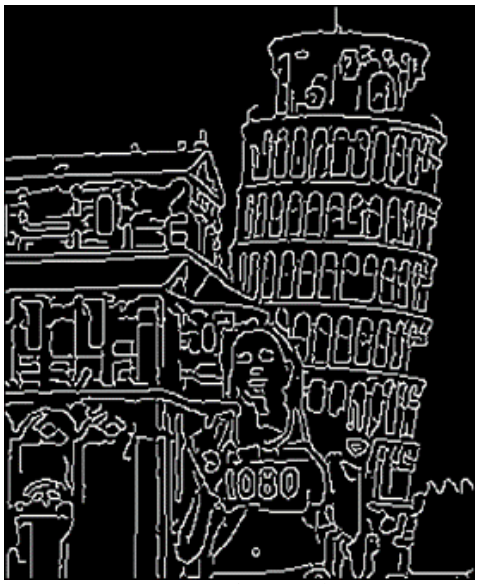

Figure 3. Edge Detection

After getting the histogram features another features are obtained by applying the canny edge detection. The edge detected in the image are shown in the figure 3.

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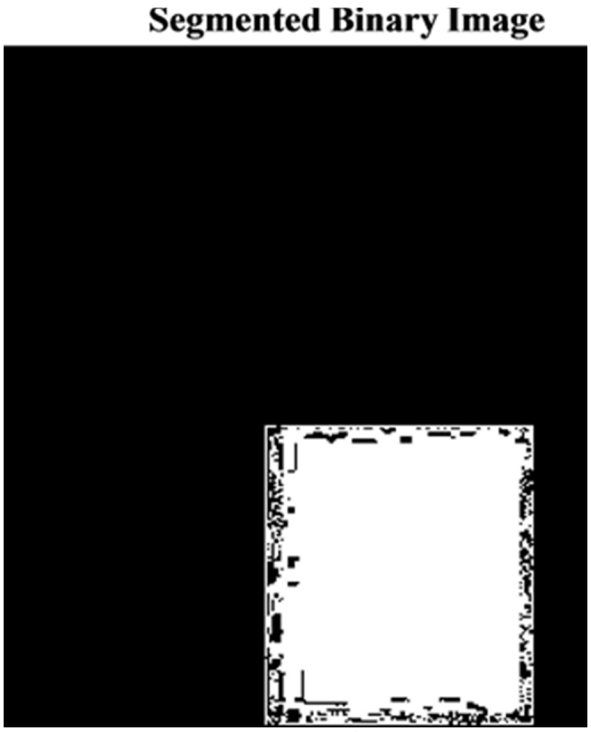

Figure 4. Segmented Image

Figure 5. Detected boundary

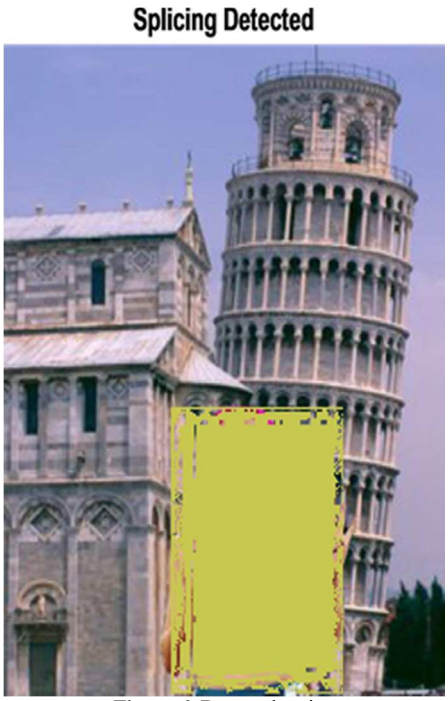

Figure 6. Detected region

Figure 6*: *Image forgery detection for image id 1 The edges are preserved as one and the image is in theblack and white format as the binary image. On theedges the image segmentation is applied and thesegment with extra copy paste part is detected as theimage region under forgery as shown in the figure 4.

The input colored image in RGB space is superimposed on the segmented image to demonstrate the forged region in the figure 5. Finally the region under copy paste image forgery is subtracted out from the input image and shown in figure 6.

TABLE I. RESULT PERFORMANCE

Precision	Recall	F-Score Measure
0.8535	1	0.9209
0.9008	1	0.9478
0.8635	1	0.9267
0.8843	1	0.9386
0.8115	1	0.8960
0.9216	1	0.9592
0.8988	1	0.9467
0.8721	1	0.9317

Similar to previous experiment the proposed detection algorithm is applied for 8 different images and finally the precision, recall and F-score is calculated (Table 1).The average F-score is 0.93

TABLE II. AVERAGE F1 SCORES OF THE BASELINE METHODS

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Method	CNN	MFCN	NOI1	DCT	CFA2	BLK	ELA	CFA1	ADQ2	NADQ	**Proposed**
F1 Score	0.7468	0.5410	0.2633	0.3005	0.2125	0.2312	0.2136	0.2073	0.3359	0.1763	**0.93**

The results are compared to existing methods F scoreto the other image forgery detection methods. It is observed that the F1bscore for proposed method is highest to other existing methods

V. CONCLUSION

The suggested approach is reliable since it uses a segmentation technique, is reliable as it is less completed and expensive because we both feature point-based and splicing-based algorithms are used. In order to match the feature and extract the tampered region, we further applied a clever edge detection method. The overall accuracy using a database of various photos is 98 percent. In our future work, we hope to build the detection method with higher recall rates as well as some other artificial intelligence tampering strategies.

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